MICHIGAN TECH · SPRING 2027

Wanted: Future Autonomy Stack Engineers.



EE5531 Introduction to Robotics

Transform **raw sensor streams** into **autonomous movement**. You will build software to map environments, estimate position under uncertainty, and navigate physical platforms using algorithmic foundations standard in industry.



Built by students: 3D LiDAR map of Husky Plaza, about 2 million points

Real robots

Clearpath Jackals and TurtleBot3s with LiDAR and IMUs. Not just simulation.

The industry toolchain

ROS 2, Nav2, SLAM, Gazebo, RViz2, Python, and Git: the skills named in robotics and AV job postings.

The math behind autonomy

Probabilistic sensor models, Bayesian filtering (EKF/UKF), mapping, and path planning.

A portfolio you can show

Public GitHub repos with code, maps, and video. Exactly what today's employers expect.

COURSE STRUCTURE: THE OODA LOOP

Observe — Sensor models, noise, and calibration

› **Orient** — State estimation, mapping, and SLAM

› **Decide** — Costmaps, path planning, and behavior trees

› **Act** — Control and full-stack integration on the Jackals

— WHERE EE5531 ALUMNI WORK NOW —

MAY mobility · TORC · gm general motors · TORO · Qualcomm

...and throughout the autonomous vehicle sector.

LECTURE	Mon & Wed, 1:00–1:50 PM
PROJECTS	Mon, 2:00–3:50 PM
CRN	12485 / 12486
PREREQS	FOSS101 or equivalent experience
FOR	MTU Robotics Engineering BS grads, and anyone with Linux, Python/C++, and Git experience.
CONTACT	Dr. Jeremy Bos · jpbos@mtu.edu

Runs only with sufficient enrollment. Register early.

Scan for the full course site

robust-autonomous-systems-laboratory.github.io/ee5531